Ibn Tofail University

Algebra IV (S3)

Problem Set II

Exercise 1:

Triangularize the following matrices in $M_3(\mathbb{R})$ when possible.

$$A = \begin{pmatrix} 2 & 0 & 4 \\ 3 & -4 & 12 \\ 1 & -2 & 5 \end{pmatrix} \quad B = \begin{pmatrix} -2 & 2 & -1 \\ -1 & 1 & -1 \\ -1 & 2 & -2 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 1 & 3 \end{pmatrix}$$

Correction

For matrix A: Characteristic polynomial:

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 2-\lambda & 0 & 4 \\ 3 & -4-\lambda & 12 \\ 1 & -2 & 5-\lambda \end{vmatrix} \\ &= (2-\lambda)[(-4-\lambda)(5-\lambda)+24] - 0 + 4[-6-(-4-\lambda)] \\ &= -\lambda^3 + 3\lambda^2 - 10\lambda + 16 \end{aligned}$$

Eigenvalues: $\lambda = 2, 2, 4$ (all real). Since all eigenvalues are real, A is triangularizable over $\mathbb{R}$.

For matrix B: Characteristic polynomial:

$$\begin{aligned} \det(B - \lambda I) &= \begin{vmatrix} -2-\lambda & 2 & -1 \\ -1 & 1-\lambda & -1 \\ -1 & 2 & -2-\lambda \end{vmatrix} \\ &= -(\lambda+1)^2(\lambda-1) \end{aligned}$$

Eigenvalues: $\lambda = -1$ (multiplicity 2), $\lambda = 1$ (all real). Since all eigenvalues are real, B is triangularizable over $\mathbb{R}$.

For matrix C: Characteristic polynomial:

$$\begin{aligned} \det(C - \lambda I) &= \begin{vmatrix} 2-\lambda & 0 & 1 \\ 1 & 1-\lambda & 0 \\ -1 & 1 & 3-\lambda \end{vmatrix} \\ &= -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = -(\lambda-1)(\lambda-2)(\lambda-3) \end{aligned}$$

Eigenvalues: $\lambda = 1, 2, 3$ (three distinct real eigenvalues). Since all eigenvalues are real, C is triangularizable over $\mathbb{R}$.

All three matrices are triangularizable over $\mathbb{R}$.

Exercise 2:

Determine the minimal polynomial of the following matrices:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 1 & -2 \\ 2 & 2 & -3 \end{pmatrix} \quad D = \begin{pmatrix} 3 & 0 & 8 \\ 3 & -1 & 6 \\ -2 & 0 & -5 \end{pmatrix}$$

Correction

For matrix A: Characteristic polynomial: $(1-\lambda)^2$
Since $(A-I)^2 = 0$ but $A - I \neq 0$, minimal polynomial is $(\lambda - 1)^2$.
For matrix B: Characteristic polynomial: $-\lambda^2(\lambda - 3)$
Since $B(B-3I) = 0$ but neither B nor $B-3I$ alone gives zero, minimal polynomial is $\lambda(\lambda - 3)$.
For matrix C: Characteristic polynomial: $-\lambda(\lambda+1)^2$
Check: $(C+I)^2 \neq 0$ but $C(C+I)^2 = 0$, so minimal polynomial is $\lambda(\lambda+1)^2$.
For matrix D: Characteristic polynomial: $-(\lambda+1)^2(\lambda-1)$
Check: $(D+I)^2(D-I) = 0$ but $(D+I)(D-I) \neq 0$, so minimal polynomial is $(\lambda+1)^2(\lambda-1)$.

Exercise 3:

Let A be the matrix:

$$A = \begin{pmatrix} 1 & 1-b & b & 0 \\ -a & a & -a & 1+b \\ -a & a-1 & 1-a & 0 \\ 1 & -b & 1 & a \end{pmatrix}$$

where $a, b \in \mathbb{R}$.

1. Give a necessary and sufficient condition for A to be diagonalizable.
2. Using the spectrum of A (the set of eigenvalues of A), show that $A + I_4$ is invertible.
3. Calculate $(A + I_4)^{-1}$ when A is diagonalizable.

Correction

1. the characteristic polynomial is:$\det(A - \lambda I) = (\lambda - a)(\lambda - 1)^3$
Eigenvalues: $\lambda = a$ (simple), $\lambda = 1$ (triple).
For diagonalizability, the geometric multiplicity of $\lambda = 1$ must equal its algebraic multiplicity (3). This occurs when $\dim(\ker(A - I)) = 4 - \operatorname{rank}(A - I) = 3$.
2. The spectrum of A is $\{a, 1\}$. Since -1 is not an eigenvalue (as $a, 1 \neq -1$), we have $\det(A + I_4) \neq 0$, so $A + I_4$ is invertible.
3. When A is diagonalizable ($a \neq 1$ and $b = 0$), we can write $A = PDP^{-1}$ where D is diagonal. Then:

$$(A + I_4)^{-1} = (PDP^{-1} + I_4)^{-1} = P(D + I_4)^{-1}P^{-1}$$

where $(D + I_4)^{-1}$ is the diagonal matrix with entries $\frac{1}{\lambda_i + 1}$.

Exercise 4:

Let $A \in M_n(\mathbb{R})$.

1. If A is invertible, show that A is diagonalizable if and only if A^2 is diagonalizable.
2. Does the result still hold if A is not invertible?

Correction

1. ($\Rightarrow$) If A is diagonalizable, then $A = PDP^{-1}$ with D diagonal. Then $A^2 = PD^2P^{-1}$, so A^2 is diagonalizable.
($\Leftarrow$) If A^2 is diagonalizable and A is invertible, then all eigenvalues of A are nonzero. Let $A^2 = QDQ^{-1}$. Since D has positive diagonal entries (squares of eigenvalues), we can take $A = Q\sqrt{D}Q^{-1}$ using the principal square root, so A is diagonalizable.
2. The result does not hold when A is not invertible. Counterexample:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Then $A^2 = 0$ (diagonalizable), but A is not diagonalizable since its minimal polynomial is λ^2.

Exercise 5:

Let E be a finite-dimensional $\mathbb{R}$-vector space and f an endomorphism of E such that:

$$(f - id_E)^3 \circ (f + 2id_E) = 0$$

$$(f - id_E)^2 \circ (f + 2id_E) \neq 0$$

Is f diagonalizable?

Correction

The given conditions tell us that:

- The minimal polynomial $m_f(\lambda)$ divides $(\lambda - 1)^3(\lambda + 2)$
- $m_f(\lambda)$ does not divide $(\lambda - 1)^2(\lambda + 2)$

Therefore, the minimal polynomial must be exactly $m_f(\lambda) = (\lambda - 1)^3(\lambda + 2)$.
Since the minimal polynomial has repeated roots $(\lambda - 1)$ with multiplicity 3, f is **not diagonalizable**.
For an endomorphism to be diagonalizable, its minimal polynomial must have only simple roots.

Exercise 6:

Let $n \in \mathbb{N}^*$ and $A \in M_n(\mathbb{R})$ such that $A^4 = 7A^3 - 12A^2$. Show that A is triangularizable in $M_n(\mathbb{R})$, that $\text{tr}(A) \in \mathbb{N}$ and that $\text{tr}(A) \leq 4n$.

Correction

Given: $A^4 = 7A^3 - 12A^2$
Thus: $A^2(A - 3I)(A - 4I) = 0$
Therefore, the minimal polynomial $m_A(\lambda)$ divides $\lambda^2(\lambda - 3)(\lambda - 4)$.
The eigenvalues of A are among $\{0, 3, 4\}$, all of which are real. Since all eigenvalues are real, A is triangularizable over $\mathbb{R}$.
Let the multiplicities be:

- $\lambda = 0$: k times.
- $\lambda = 3$: m times.
- $\lambda = 4$: p times.

with $k + m + p = n$
Then: $\operatorname{tr}(A) = 0 \cdot k + 3m + 4p = 3m + 4p$
Since $m, p \in \mathbb{N}$ and $m + p = n - k \leq n$, we have:

- $\operatorname{tr}(A) = 3m + 4p \in \mathbb{N}$
- $\operatorname{tr}(A) = 3m + 4p \leq 3(m + p) + p = 3(n - k) + p \leq 3n + p \leq 4n$

Exercise 7:

Let $n \in \mathbb{N} \setminus \{0, 1\}$ and $A = (a_{ij}) \in M_n(\mathbb{R})$ with $a_{1j} = a_{j1} = 1$ for $j > 1$ and $a_{11} = a_{ij} = 0$ for $i, j > 1$.

1. Calculate A^3 and deduce a simple relation between A^3 and A.
2. Deduce that A is diagonalizable and give its eigenvalues.
3. Diagonalize A.

Correction

The matrix has the form:

$$A = \begin{pmatrix} 0 & 1 & 1 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

1. Compute A^2:

- $(A^2)_{11} = \sum_{j=2}^{n} 1 \cdot 1 = n - 1$
- $(A^2)_{1j} = 0$ for $j > 1$
- $(A^2)_{i1} = 0$ for $i > 1$
- $(A^2)_{ij} = 1$ for $i, j > 1$

Now compute $A^3 = A \cdot A^2$:

- $(A^3)_{11} = 0$
- $(A^3)_{1j} = n - 2$ for $j > 1$
- $(A^3)_{i1} = n - 2$ for $i > 1$
- $(A^3)_{ij} = 0$ for $i, j > 1$

Thus $A^3 = (n-2)A$
2. The relation $A^3 = (n-2)A$ implies the minimal polynomial divides

$$\lambda(\lambda^2 - (n-2)) = \lambda(\lambda - \sqrt{n-2})(\lambda + \sqrt{n-2})$$

.
Since the minimal polynomial has distinct roots, A is diagonalizable.
Eigenvalues: 0, $\sqrt{n-2}$, $-\sqrt{n-2}$
3. The eigenspaces:

- For $\lambda = 0$: vectors with $x_1 = 0$ and $\sum_{i=2}^{n} x_i = 0$ (dimension $n-2$)
- For $\lambda = \sqrt{n-2}$: vector $(\sqrt{n-2}, 1, 1, \ldots, 1)$
- For $\lambda = -\sqrt{n-2}$: vector $(-\sqrt{n-2}, 1, 1, \ldots, 1)$

Exercise 8:

Let $A \in M_3(\mathbb{R})$ such that $A^3 + A = 0$ and $A \neq 0$.

1. Determine spect(A) (the spectrum of A).
2. Is A diagonalizable in $M_3(\mathbb{R})$?
3. Is A diagonalizable in $M_3(\mathbb{C})$?
4. Show that A is similar to: $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$ in $M_3(\mathbb{R})$.

Correction

Given: $A^3 + A = 0$ and $A \neq 0$
Rewriting: $A(A^2 + I) = 0$
1. The characteristic polynomial divides $\lambda(\lambda^2 + 1)$, so the eigenvalues are among $\{0, i, -i\}$.
Since $A \neq 0$, at least one eigenvalue is nonzero. The possible spectra over $\mathbb{C}$ are:

- $\{0, i, -i\}$ (all three eigenvalues)
- $\{i, -i\}$ (with one missing, but sum must be real)
- $\{0\}$ is impossible since $A \neq 0$

Thus $\text{spect}(A) = \{0, i, -i\}$ over $\mathbb{C}$, and $\text{spect}_{\mathbb{R}}(A) = \{0\}$.
2. No, A is not diagonalizable over $\mathbb{R}$ because it has complex eigenvalues i and $-i$ that are not real.
3. Over $\mathbb{C}$, if the geometric multiplicity equals algebraic multiplicity for each eigenvalue, then A is diagonalizable. Since all eigenvalues are distinct, A is diagonalizable over $\mathbb{C}$.
4. The matrix $B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$ satisfies $B^3 + B = 0$ and has the same minimal polynomial $\lambda(\lambda^2+1)$ as A. Since both matrices have the same rational canonical form, they are similar over $\mathbb{R}$.

Exercise 9:

Let $A = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$ where $n \in \mathbb{N}^*$.

1. Calculate A^2.
2. Is A diagonalizable:
   (a) In $M_{2n}(\mathbb{R})$?
   (b) In $M_{2n}(\mathbb{C})$?

   If yes, perform the diagonalization.

Correction

1. Compute A^2:

$$A^2 = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} = \begin{pmatrix} -I_n & 0 \\ 0 & -I_n \end{pmatrix} = -I_{2n}$$

2.

1. Over $\mathbb{R}$: The minimal polynomial is λ^2+1, which has no real roots. Therefore, A is **not diagonalizable** over $\mathbb{R}$.
2. Over $\mathbb{C}$: The minimal polynomial $\lambda^2 + 1 = (\lambda - i)(\lambda + i)$ has distinct roots, so A is **diagonalizable** over $\mathbb{C}$.

   To diagonalize, find eigenvectors:

   - For $\lambda = i$: Solve $(A - iI)v = 0$
   - For $\lambda = -i$: Solve $(A + iI)v = 0$

   Let P be the matrix whose columns are eigenvectors, then $A = PDP^{-1}$ where $D = \text{diag}(i, i, \dots, i, -i, -i, \dots, -i)$.

Exercise 10:

Let E be a finite-dimensional $\mathbb{R}$-vector space and f an endomorphism of E such that $f^2 = f$ and $f \neq 0$. How should t be chosen so that $(id_E + tf)$ is invertible? Calculate $(id_E + tf)^{-1}$.

Correction

Given: $f^2 = f$ (so f is a projection) and $f \neq 0$.
The eigenvalues of f are 0 and 1 (since $f^2 = f$ implies $\lambda^2 = \lambda$).
Then the eigenvalues of $id_E + tf$ are $1 + t \cdot 0 = 1$ and $1 + t \cdot 1 = 1 + t$.
For $id_E + tf$ to be invertible, all eigenvalues must be nonzero:

1. $1 \neq 0$ (always true)

2. $1 + t \neq 0$

Thus $id_E + tf$ is invertible if and only if $t \neq -1$.
To find the inverse, guess it has the form $id_E + \alpha f$ for some $\alpha \in \mathbb{R}$.
Compute: $(id_E + tf)(id_E + \alpha f) = id_E + \alpha f + tf + t\alpha f^2$
Since $f^2 = f$, this equals: $id_E + (\alpha + t + t\alpha)f$
We want this to equal id_E, so we need: $\alpha + t + t\alpha = 0 \Rightarrow \alpha(1+t) = -t \Rightarrow \alpha = -\frac{t}{1+t}$
Thus: $(id_E + tf)^{-1} = id_E - \frac{t}{1+t} f$
Verification: $(id_E + tf)\left(id_E - \frac{t}{1+t} f\right) = id_E - \frac{t}{1+t} f + tf - \frac{t^2}{1+t} f = id_E$